

Capstone Project Proposal – Step 2

BindFlow – Digital Bookbinding Production & Order Management System

Stack

- **Frontend:** React.js (with React Router and Axios)
- **Backend:** Node.js + Express.js
- **Database:** MongoDB (with Mongoose ODM)
- **Authentication:** JWT (JSON Web Tokens) + bcrypt for password hashing
- **Other Tools:** Google Books API / Open Library API, Cloudinary (for file uploads), jsPDF (for quote/invoice generation)

Focus

This will be an **evenly focused full-stack application**. I will put strong effort into both frontend (clean, intuitive UI/UX with dashboards) and backend (complex data relationships, workflows, business logic, and secure routes).

Type

Website (responsive web application that works well on desktop and mobile browsers).

Goal

The goal of BindFlow is to digitize and streamline the current manual, paper-based processes of a bookbinding business. It will replace three physical notebooks (Orders, Quotes, and Maquila/Production book) with a centralized digital system. This will reduce errors, save time on WhatsApp quote requests, improve production tracking, and give clients real-time visibility into their job status.

Users

- **Primary Users (Internal):** Business owners, sales staff, production supervisors, and managers (ages 25–55).
- **Secondary Users (External):** Clients (printing shops, schools, individuals) who submit orders and track their jobs (ages 20–50).

Data

I will use a combination of:

- **External API:** Google Books API or Open Library API to automatically fetch book title, author, ISBN, and cover image when clients enter a book.
- **Custom Data:** All business-specific data (quotes, orders, production status, payments, deliveries) will be stored in my own MongoDB database.
- Data will be collected through user input forms (order submission, quote creation, status updates) and API calls.

Breaking Down the Project

Database Schema (High-Level)

- **User** (role: client, staff, admin)
- **Order** (client info, status, created date)
- **Quote** (linked to Order, pricing details, specifications)
- **Job/Maquila** (production details, partial quantities, delivery dates, current status)
- **Payment** (amount, status, verification notes)
- **Delivery** (multiple partial deliveries per job)

Key Functionality & User Flow

1. **Client Flow:**
Register/Login → Search book via API → Fill binding specifications → Submit order → Track real-time status.
2. **Staff Flow:**
Login to dashboard → View new orders → Create/approve quotes → Update production status → Mark partial deliveries → Generate PDF quotes.

Features (Must-Have vs Stretch)

Must-Have Features (Core for Capstone):

- User authentication with different roles
- Client order submission with Google Books API integration
- Admin/staff dashboard to view and manage all orders
- Quote creation and pricing logic
- Job status tracking (Pending, In Production, Ready, Delivered)
- Basic search and filter functionality
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Stretch Goals (Nice-to-have):

- PDF quote/invoice generation (jsPDF)
- Digital signature capture for job pickup
- Simple analytics dashboard (charts with Recharts)
- File upload for book covers or reference images (Cloudinary)
- Real-time status updates using Socket.io (if time allows)

Potential Issues & Solutions

- API limitations (Google Books): Will have fallback manual entry form.
- Complex workflows: Will start with simple status updates and expand.
- Sensitive data: Client information and pricing will be secured with proper authentication and authorization middleware.
- No web scraping will be used.

What Makes This More Than a Basic CRUD App

- Multiple user roles with different permissions
- Real business logic for quote calculation
- Multi-step workflow (Order → Quote → Production → Delivery → Payment)
- Integration of external API with custom business rules
- Real-world impact on my family business

Task Breakdown

Task Name	Description	Type	Difficulty	Label
Design Database Schema	Create models for User, Order, Quote, Job, etc.	Backend	Medium	Must Have
Source Data & API	Integrate Google Books API + manual entry	Fullstack	Medium	Must Have
User Authentication	Register, Login, Role-based access	Fullstack	Medium	Must Have
Client Order Submission	Form with API lookup	Fullstack	Medium	Must Have
Admin Dashboard	View, manage orders and statuses	Fullstack	Hard	Must Have
Quote Generation	Pricing logic + display	Backend	Medium	Must Have
PDF Generation	Generate quotes as PDF	Frontend	Easy	Stretch Goal
Digital Signature	Capture signature on delivery	Fullstack	Medium	Stretch Goal